Leverage Direction Matters: Cross-Asset Evidence on GARCH Model Selection and Volatility Targeting*

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May 2026 (v3.3)

Abstract

The direction of the asymmetric volatility response—measured by the GJR-GARCH γ parameter—varies systematically across asset classes in ways that have direct implications for model selection, risk management, and portfolio construction. Using daily data for seven primary assets (equities, gold, bonds, emerging markets, cryptocurrency) over 2017–2025 (with 2026 reserved for out-of-sample validation), extended to 26 assets in validation analyses, we make two contributions. First, we propose a leverage direction taxonomy: equities exhibit standard leverage ($\gamma > 0$), gold displays regime-dependent leverage (inverted during fear-driven rallies, standard during liquidation-driven declines), and bonds show near-zero asymmetry. This taxonomy determines optimal GARCH specification: GJR-GARCH significantly outperforms symmetric GARCH only for assets with statistically significant leverage ($t > 1.65$ on the quarterly-mean HAC statistic), and the rule-prescribed model is never significantly beaten across the nine Diebold-Mariano comparisons in the primary sample (in-sample threshold calibration), with 6/6 correct out-of-sample predictions. Second, we link leverage direction to the economic channel of volatility targeting (VT) alpha: γ predicts whether VT acts as trend-following or contrarian within homogeneous equity markets (Spearman $\rho = 0.886$, $p = 0.019$

*I thank seminar participants for helpful comments. All errors remain my own. Data and replication code available upon request.

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for six equity-type assets), but this relationship does not extend to diverse asset classes ($\rho = -0.448$, $p = 0.14$ for twelve assets). Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across all tested assets and driven by base volatility level ($\rho = 0.944$ for primary assets, $\rho = 0.83$ for the extended $N = 14$ sample), independent of γ. These findings delineate a complexity ceiling: increasing model sophistication (DCC-GARCH, copula, GARCH-MIDAS, Markov-switching) improves statistical measurement but not investable allocation decisions.

Keywords: GARCH, leverage effect, gold, asymmetric volatility, Value-at-Risk, Basel III, volatility targeting, cross-asset, regime-dependent

JEL Classification: C22, C53, G01, G11, G17

1 Introduction

The GARCH family of models (Bollerslev, 1986) remains the workhorse for daily volatility forecasting in financial markets. A substantial literature has developed around asymmetric extensions—most notably the GJR-GARCH of Glosten et al. (1993) and the EGARCH of Nelson (1991)—designed to capture the empirical observation that negative equity returns tend to increase future volatility more than positive returns of equal magnitude. This "leverage effect," first noted by Black (1976) and formalized by Christie (1982), is well-documented for equity markets and has become a standard consideration in volatility model specification.

However, the implicit assumption in much of the applied GARCH literature—that asymmetric models uniformly improve upon their symmetric counterparts—warrants closer examination when moving beyond equities. While Chevallier and Ielpo (2017) document inverted asymmetric volatility in gold and several agricultural commodities, the implications of this reversal for model selection, risk management, and portfolio construction have not been systematically explored.

This paper makes two contributions to the volatility forecasting and financial risk management literature.

First, we propose a leverage direction taxonomy based on the GJR-GARCH γ parameter across seven primary assets spanning equities (SPY, QQQ, EEM), precious metals (GLD, SLV), government bonds (TLT), and cryptocurrency (BTC-USD), validated on up to 26 assets. The sign of γ corresponds to the asset's price-driving mechanism: risk assets exhibit standard leverage ($\gamma > 0$), safe-haven assets show inverted leverage ($\gamma < 0$), and interest rate instruments display no significant asymmetry ($\gamma \approx 0$). Gold's inverted leverage is *regime-dependent* rather than unconditional: the canonical full-replication estimate of the rolling-window mean is statistically indistinguishable from zero (mean $\gamma = +0.002$, HAC $t = +0.15$, with 67% of quarterly windows negative; K903), while the regime decomposition shows inversion concentrated in fear-driven rallies but standard leverage during liquidation-driven declines ($t = -3.79$, $p < 0.001$ for the regime difference). This taxonomy determines optimal GARCH specification: GJR-GARCH significantly outperforms symmetric GARCH only for assets with statistically significant asymmetry; within our sample, any magnitude cutoff in the band $|\gamma| \in (0.009, 0.072)$ yields the same classification as the significance rule (see Section 4.2.2). Under the t-statistic rule ($t > 1.65$ on the quarterly-mean HAC statistic), the prescribed model is never significantly beaten in any of the nine Diebold-Mariano comparisons in the *primary* sample (in-sample, Table 4); BTC-USD illustrates that a significant γ ($t = +2.88$) does not guarantee a significant forecasting edge—its two models are statistically indistinguishable out of sample. The rule also classifies 6/6 comparisons correctly out-of-sample, using 2023 γ estimates to predict 2024–2025 model choice. We propose that γ direction, rather than the commonly used return skewness, provides a more reliable model selection criterion.

Second, we apply the leverage taxonomy to volatility targeting (VT), linking leverage direction to the economic channel through which VT generates alpha. This generalizes the equity-specific finding of Hood and Raughtigan (2025). Within equity-type assets, γ predicts whether VT acts as trend-following or contrarian (Spearman $\rho = 0.886$, $p = 0.019$ for six equity-type assets; see Section 5.3 for domain restrictions), with out-of-sample predictive power ($\rho = 0.821$ using γ from 2010–2017 to predict trend beta in 2018–2026). However, this relationship does not extend to diverse asset classes ($\rho = -0.448$,

$p = 0.14$ for twelve assets), delineating the proposition's domain. Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across all tested assets and driven by base volatility level (Pearson $\rho = 0.944$, $p = 0.016$, $N = 5$ primary assets; Pearson $\rho = 0.83$, $p = 0.0002$, $N = 14$ in the extended sample), independent of γ.

Supporting findings reinforce a broader pattern: increasing model sophistication improves statistical measurement but not investable allocation decisions. Supplementary evidence on cross-market transmission and auxiliary robustness checks is reported in the online supplement.

The remainder of the paper is organized as follows. Section 2 reviews the literature. Section 3 describes data and methodology. Section 4 presents empirical results. Section 5 discusses implications and limitations. Section 6 concludes.

2 Literature Review

The paper sits at the intersection of three literatures. The first studies asymmetric volatility models: GARCH captures volatility clustering, while GJR-GARCH and EGARCH allow negative and positive shocks to affect variance differently (Bollerslev, 1986; Glosten et al., 1993; Nelson, 1991). In equities, this leverage effect is standard and economically tied to balance-sheet amplification (Black, 1976; Christie, 1982; Engle and Siriwardane, 2018). A smaller literature shows that non-equity assets can reverse the sign of the asymmetry, especially for gold and some commodities, but stops short of turning that sign information into a model-selection rule (Chevallier and Ielpo, 2017; Batten et al., 2010; Chang et al., 2021).

The second literature concerns risk management. Basel backtesting makes the distributional assumption as important as the variance equation itself (BCBS, 2006, 2019; Kupiec, 1995; Christoffersen, 1998). Student-t and skewed-t innovations often improve tail coverage relative to Gaussian specifications (Bollerslev, 1987; Hansen, 1994; Kuester et al., 2006). The third literature studies volatility targeting: inverse-volatility scaling improves drawdown control and sometimes improves Sharpe ratios, though the evidence is criterion- and asset-dependent (Fleming et al., 2001, 2003; Moreira and Muir, 2017; Harvey et al., 2018; Cederburg et al., 2020; Bozovic, 2024; DeMiguel et al., 2024; Hood and Raughtigan, 2025; Xu, 2024). Our contribution is to connect these strands through a single empirical object, the sign and magnitude of γ, and then test where that information matters for forecasting, VaR, and portfolio construction.

3 Data and Methodology

3.1 Data

We use daily adjusted closing prices for seven exchange-traded assets: SPDR S&P 500 ETF Trust (SPY), Invesco QQQ Trust (QQQ), iShares MSCI Emerging Markets ETF (EEM), SPDR Gold Shares (GLD), iShares Silver Trust (SLV), iShares 20+ Year Treasury Bond ETF (TLT), and Bitcoin (BTC-USD). Data are sourced from Yahoo Finance adjusted close prices for the period January 2017 through March 2026.

Data source justification. Yahoo Finance provides the daily history for all assets. This choice is standard in empirical asset-pricing practice and is defensible here because the sample consists of large, liquid ETFs and Bitcoin, not microcaps where stale prices and delayed corporate-action adjustments are more problematic (Bali et al., 2016). We validate the feed by spot-checking crisis dates against Bloomberg, matching Yahoo VIX to official series, and confirming that return moments line up with overlapping benchmark studies (Moreira and Muir, 2017; Cederburg et al., 2020). Daily returns are computed as simple percentage changes. We nevertheless treat institutional-grade replication as the relevant production standard.

Table 2 reports descriptive statistics. All return series reject normality (Jarque-Bera $p < 0.001$), exhibit significant ARCH effects (Engle's LM test (Engle, 1982), $p < 0.001$), and are stationary (ADF $p < 0.001$). Excess kurtosis ranges from 3.52 (GLD) to 14.61 (SPY), motivating the consideration of fat-tailed distributions for VaR estimation.

Sample period justification. The primary sample spans January 2017 to March 2026, with 2023–2024 as the main out-of-sample period and 2025–March 2026 as validation. This is shorter than some long-horizon GARCH studies, but each forecast is estimated on a rolling two-year window, and robustness checks extend the effective estimation history with larger windows back to 2006. The key object is not long-horizon return predictability but the sign and usefulness of γ, which is stable across the window sizes we test and remains qualitatively unchanged in stressed subperiods such as 2020–2022 and the 2025–2026 Iran/Hormuz episode.

3.2 Volatility Models

We compare the symmetric GARCH(1,1) of Bollerslev (1986), $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, against the GJR-GARCH(1,1) of Glosten et al. (1993):

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbf{1}_{\varepsilon_{t-1}<0})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{1}$$

where γ captures asymmetric responses: $\gamma > 0$ implies standard leverage (negative returns increase variance), $\gamma < 0$ implies inverted leverage.

Mean specification. We adopt the zero-mean specification $r_t = \sigma_t\varepsilon_t$ throughout. This choice is grounded in the well-documented finding that the conditional mean of daily equity returns is negligible relative to conditional variance: for a typical daily mean of 0.04% and daily standard deviation of 1%, the mean-to-volatility ratio is approximately 0.04, making its contribution to variance dynamics immaterial (Bollerslev, 1986; Hansen and Lunde, 2005). Francq and Zakoïan (2004) show that quasi-maximum likelihood estimation of GARCH parameters remains consistent regardless of mean specification when the true conditional mean is small relative to conditional variance. As a robustness check, we re-estimate all models with an AR(1) mean specification $r_t = \mu + \phi r_{t-1} + \sigma_t\varepsilon_t$; results are qualitatively unchanged—QLIKE rankings, gamma sign classifications, and DM test conclusions are all preserved (Section 4.6, Model Specification).

All models are estimated with Normal quasi-likelihood via the `arch` package (Sheppard, 2023). For VaR, we additionally consider Student-t innovations with fixed degrees of freedom.

3.3 Rolling Estimation

We employ a rolling window approach with re-estimation at each forecast origin. The primary window size is $w = 504$ trading days (approximately 2 calendar years), which satisfies the minimum sample size recommendation for GARCH estimation (≥ 500 observations; see Hwang and Valls Pereira 2006) while avoiding contamination from distant regime changes. Robustness checks with $w \in \{252, 1000, 2000, 3000, 5000\}$ (reported in the online supplement) confirm that QLIKE rankings are qualitatively robust to window choice, though $w = 504$ produces the best QLIKE during high-volatility periods (2020–2021) while larger windows ($w \geq 2000$) perform marginally better during calm periods (2023–2024). The gamma sign classification is invariant to window size across all tested values.

For each out-of-sample date t, we estimate the model using data from $t - w$ to $t - 1$, and produce a one-step-ahead variance forecast $\hat{\sigma}_t^2$. The realized variance proxy is the squared daily return r_t^2.

3.4 Evaluation Criteria

3.4.1 Statistical Loss Functions

The primary evaluation metric is the QLIKE loss function (Patton, 2011), defined as the Gaussian quasi-log-likelihood contribution:[1]

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^{T} \left(\frac{r_t^2}{\hat{\sigma}_t^2} + \ln \hat{\sigma}_t^2 \right) \tag{2}$$

[1] This formulation is equivalent to $-\ln f(r_t \mid \hat{\sigma}_t^2)$ under Gaussian quasi-likelihood, up to an additive constant. Values are negative for daily returns because $\ln \hat{\sigma}_t^2 \approx \ln(10^{-4}) \approx -9.2$. **Lower (more negative) values indicate better forecasts.** An alternative convention in the literature defines the Patton (2011) centered loss as $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which produces values centered near 1.0. Both formulations preserve forecast rankings; we adopt the quasi-log-likelihood form throughout the main text and the online supplement for consistency with maximum likelihood estimation.

QLIKE is robust to noise in the realized variance proxy and ranks forecasts consistently regardless of the proxy used, provided the proxy has equal bias across models.

3.4.2 Diebold-Mariano Test

We test the null hypothesis of equal predictive accuracy between models using the Diebold-Mariano test (Diebold and Mariano, 1995) with Newey-West HAC standard errors (5 lags):

$$DM = \frac{\bar{d}}{SE_{NW}(\bar{d})} \sim N(0, 1) \tag{3}$$

where $d_t = L(r_t^2, \hat{\sigma}_{1,t}^2) - L(r_t^2, \hat{\sigma}_{2,t}^2)$ is the loss differential using QLIKE.

3.4.3 VaR Backtesting

Value-at-Risk at confidence level α is computed as:

- **Normal:** $\text{VaR}_\alpha = \Phi^{-1}(\alpha) \cdot \hat{\sigma}_t$
- **Student-t:** $\text{VaR}_\alpha = t_\nu^{-1}(\alpha) \cdot \sqrt{(\nu - 2)/\nu} \cdot \hat{\sigma}_t$

where ν is the degrees of freedom (fixed at 5 in our baseline, consistent with typical empirical estimates for financial returns). The choice of df = 5 is validated out-of-sample: Section 4.4 demonstrates that Basel III Green Zone compliance is achieved for all df $\leq$ 6, and that jointly estimating df produces *worse* performance by over-adapting to quiet markets. The fixed df = 5 is thus a conservative choice that is robust across the empirical range df $\in$ [4, 7].

We apply Kupiec's (1995) unconditional coverage test (LR statistic, χ^2 with 1 d.f.) and Christoffersen's (1998) independence test to assess violation clustering. Annual violation counts are classified per the Basel III traffic light system (Green: 0–4, Yellow: 5–9, Red: $\geq$10 violations per approximately 250 trading days).

3.4.4 Leverage Direction Analysis

To analyze the temporal stability of the leverage direction, we estimate GJR-GARCH on semi-overlapping quarterly windows (504-day windows advanced by 63 trading days) and collect the gamma estimates. We test:

- **H0:** $E[\gamma] \geq 0$ vs. **H1:** $E[\gamma] < 0$ (inverted leverage)

using a one-sided t-test on the quarterly gamma series.

3.5 Volatility Targeting

Following Moreira and Muir (2017), the volatility-managed portfolio weight is:

$$w_t = \frac{\sigma_{\text{target}}}{\hat{\sigma}_t} \tag{4}$$

We set $\sigma_{\text{target}} = 10\%$ annualized for the GARCH-based analysis, and $\sigma_{\text{target}} = 12\%$ for the VIX-based analysis (Section 5.5), reflecting the VIX's systematic upward bias from the embedded variance risk premium. Weights are smoothed with a 5-day moving average and clipped to $[0, 1.5]$. Post-hoc transaction cost analysis confirms that the estimated annual turnover of 756% generates cost drag of only 8–15 basis points at typical SPY bid-ask spreads.

4 Empirical Results

4.1 Data Characteristics

Table 2 summarizes the distributional properties of daily returns for our seven assets over the in-sample period 2017–2025 (with 2026 reserved for out-of-sample validation). All return series exhibit significant excess kurtosis, ranging from 3.52 (GLD) to 14.61 (SPY), and decisively reject normality (Jarque-Bera $p < 0.001$). The ARCH LM test confirms significant conditional heteroskedasticity in all series ($p < 0.001$), justifying GARCH-family models.

Return skewness varies across assets: SPY (-0.315), QQQ (-0.189), GLD (-0.296), and EEM (-0.584) show negative skewness, while TLT ($+0.155$) is approximately symmetric. A naive model selection approach based on skewness would suggest asymmetric GARCH for all negatively skewed assets. As we demonstrate in Section 4.2, this criterion is misleading for gold: despite negative skewness, gold's conditional variance responds asymmetrically to *positive* shocks.

Annualized volatility spans an order of magnitude, from 14.6% (GLD) to 57.3% (BTC-USD), motivating our analysis of VT effectiveness as a function of base volatility (Section 4.5). The extreme minimum returns—SPY at −10.94% (March 2020 COVID crash) and BTC-USD at −37.17%—underscore the importance of fat-tailed distributions for VaR estimation (Section 4.4).

4.2 Leverage Direction Across Asset Classes

4.2.1 Rolling Gamma Estimates

We estimate the asymmetry parameter γ of GJR-GARCH(1,1) using a rolling window of 504 trading days, updating quarterly. Table 3 reports cross-sectional statistics for each asset. Figure 1 illustrates the time-series evolution of these rolling estimates across assets.

The most striking finding is the **consistent sign difference across asset classes**. For equities (SPY, QQQ), γ is uniformly positive across all quarterly estimates, indicating the well-documented leverage effect. Gold (GLD) is the most subtle case: the canonical replication (K903; $n = 58$ quarterly rolling windows, $w = 504$, 2010–2026, HAC 8 lags) yields a mean rolling γ of +0.002 (HAC $t = +0.15$), statistically indistinguishable from zero, with 67% of quarterly windows negative.[2] The unconditional mean therefore does *not* establish inverted leverage for gold; rather, the inversion is **regime-dependent**—concentrated in fear-driven rally periods and reversing to standard leverage in liquidation-driven declines ($t = -3.79$, $p < 0.001$ for the regime difference; Section 5.3). Conditional on regime, the inverted-leverage channel—where *positive* returns increase conditional variance—has not been documented in the prior GARCH literature as a systematic phenomenon.

Treasury bonds (TLT) show near-zero γ (mean = −0.005, std = 0.044, 69% of win-

[2] An earlier draft reported mean $\gamma = -0.067$ (HAC $t = -5.79$, 93% negative windows) for the same specification; the canonical re-estimation (K903, documented in `k903_vs_paper_diff.md`) reverses the sign of the mean to +0.002 and renders it insignificant. We adopt the K903 values throughout and treat the unconditional rolling-mean evidence as inconclusive; the economically interpretable result is the regime decomposition below.

dows negative), indicating no significant asymmetry. Bitcoin (BTC-USD) displays mild but statistically significant standard leverage (mean $\gamma = +0.072$, HAC $t = +2.88$), though with higher cross-window instability (std = 0.105, 25% negative windows) and occasional sign reversals.

The inverted leverage in gold is consistent with gold's role as a safe-haven asset (Baur and McDermott, 2010; Baur and Lucey, 2010). During market stress, fear-driven buying drives prices upward, creating heightened uncertainty that elevates conditional variance. Crucially, this mechanism operates opposite to equity leverage, where falling prices increase debt-to-equity ratios and firm risk (Black, 1976; Christie, 1982). The absence of asymmetry in bonds is economically intuitive: interest rate changes do not systematically alter sovereign credit risk in either direction.

4.2.2 Implications for Model Selection

Table 4 presents GARCH(1,1) versus GJR-GARCH(1,1) QLIKE comparisons across all asset-period combinations. The results reveal a clear pattern: GJR-GARCH significantly outperforms only when γ is consistently positive and above approximately 0.10.

For SPY ($\gamma \approx +0.13$), GJR achieves significantly lower QLIKE (DM $p = 0.003$ for 2023–2024, $p = 0.048$ for 2025). For QQQ ($\gamma \approx +0.12$), GJR's advantage is directionally consistent but not significant ($p = 0.181$ for 2023–2024, $p = 0.086$ for 2025). For GLD—whose mean rolling γ is statistically zero—symmetric GARCH *significantly outperforms* GJR in 2023–2024 ($\Delta = +0.39\%$, $p = 0.001$), with the same direction in 2025 ($p = 0.070$): imposing an asymmetry term on an asset without stable asymmetry actively hurts forecasts. For TLT ($\gamma \approx 0$), EEM, and BTC-USD, Diebold-Mariano tests fail to reject equal accuracy at the 5% level.

This finding has practical implications: the conventional approach of selecting asymmetric GARCH models based on return skewness can be misleading. GLD exhibits substantial negative skewness (−0.296) yet has inverted leverage—skewness-based selection would incorrectly prescribe GJR. We propose that **the sign and magnitude of estimated** γ should replace skewness as the primary criterion.

4.2.3 Robustness

The γ sign classification is robust to window size (results qualitatively identical with $w = 252$ and $w = 756$), estimation period (2007–2026 for SPY shows the same pattern), and model specification (EGARCH leverage parameter shows consistent sign alignment with GJR γ).

Because our quarterly gamma estimates use overlapping windows (504-day windows stepped by 63 days, yielding 87.5% overlap between consecutive estimates), serial dependence in the gamma series is a first-order concern. As Harri and Brorsen (2009) emphasize, overlapping observations inflate the effective sample size relative to the number of independent data points, biasing naive t-statistics upward. We address this in two ways.

First, we employ Newey-West HAC standard errors (Newey and West, 1987) with 8 lags (chosen as $\lceil 0.75 \times T^{1/3} \rceil$ where T is the number of quarterly gamma estimates), which accounts for the moving-average structure induced by the overlap. All t-statistics in Table 3 are HAC-corrected; for SPY, the corrected statistic remains strongly significant ($t = +11.08$), and for GLD the corrected statistic ($t = +0.15$) confirms that the unconditional mean carries no significance to begin with.

Second, and most importantly, we repeat the cross-sectional analysis using strictly non-overlapping 5-year windows (2006–2010, 2011–2015, 2016–2020, 2021–2025). While the smaller number of cross-sections ($N = 4$) limits formal inference, the gamma rankings for the unambiguous classes are qualitatively preserved: SPY exhibits positive mean γ in all four windows (range: $+0.14$ to $+0.27$) and TLT remains near zero in all windows ($|\bar{\gamma}| < 0.02$). Gold's non-overlapping window means were negative in an earlier-draft vintage (range -0.092 to -0.031); given the K903 sign reversal of the quarterly-mean estimate, we treat gold's unconditional sign as unresolved at the window level and rest the gold claim entirely on the regime decomposition of Section 5.3.

We conclude that while the 87.5% overlap in our primary analysis inflates the number of gamma observations, it does not drive the qualitative findings for equities and bonds. The overlap provides finer temporal resolution for regime analysis (Section 5.3).

Within the K903 canonical 2010–2026 window, GLD's γ is negative in 67% of quarterly estimates. An extended OOS validation (January 2025–March 2026, $N = 300$) confirms the equity-side findings: the gamma taxonomy is consistent across the seven assets, GJR outperforms for SPY with widening improvement (DM = 4.82, $p < 0.0001$), and 12/VIX reduces MDD by 43% during the April 2025 selloff (VIX = 52.3).

The QLIKE ranking is robust to volatility proxy choice. While short-sample comparisons ($N = 42$) using 5-minute realized variance or Parkinson range-based estimators occasionally reverse the GJR–GARCH ranking, a formal test over $N = 563$ trading days confirms GJR outperforms under both r^2 (DM $t = 3.52$) and Parkinson (DM $t = 6.39$) proxies. The short-sample reversals arise from a scale mismatch: range-based proxies are systematically lower, penalizing models forecasting higher variance. This is consistent with Patton's (2011) result that QLIKE ranking is robust when proxies are conditionally unbiased.

Extending the analysis to 2005–2016 reveals important nuance: during the 2013–2015 gold bear market, γ turned sharply positive (+0.17 to +0.30), exhibiting standard leverage akin to equities. Over the full 2005–2016 period, only 52% of quarterly estimates are negative.

This regime-dependence has a clear economic interpretation: during fear-driven gold rallies, the safe-haven mechanism produces inverted leverage (price up, uncertainty up). During gold bear markets driven by liquidation, falling gold prices increase downside risk—the same mechanism as equity leverage. This strengthens our recommendation to use the **current estimated** γ rather than fixed asset-class assignments, as even gold's leverage direction can reverse with market regime.

4.2.4 Regime-Dependent Leverage in Gold

Examining the full 2005–2026 sample reveals gold's γ is regime-dependent. During the 2013–2015 gold bear market specifically, the mean of rolling-window γ estimates *within that sub-period* rises to +0.20, exhibiting standard leverage identical to equities.[3] Inde-

[3] This sub-period mean is computed from rolling estimation windows falling entirely within 2013–2015 and is sensitive to exact window-boundary choices; see the window-boundary sensitivity note in the

pendently, partitioning the full 2005–2026 sample into bull and bear regimes and pooling within each group yields a distinct pair of estimates: bull $\gamma = -0.043$ versus bear $\gamma = +0.048$ ($t = -3.79$, $p < 0.001$). The two values (+0.20 and +0.048) reflect different statistical objects—a narrow sub-period window mean versus a full-sample binary-partition group mean—and are mutually consistent. This regime-dependence is unique to gold; SPY shows no dependence (bull $\gamma = +0.24$, bear $\gamma = +0.24$, $p = 0.95$), consistent with the capital structure mechanism operating regardless of market direction.

This finding has three implications. First, it strengthens the case for using the **current estimated** γ rather than fixed asset-class assignments. Second, the trailing 252-day return serves as a useful regime indicator, correctly predicting next-quarter γ sign in 72% of cases. Third, the regime-dependence connects the leverage effect literature to the safe-haven literature: leverage direction reflects the economic mechanism driving price changes, not merely statistical properties of the return series.

4.3 GARCH vs. GJR-GARCH: Forecasting Comparison

4.3.1 QLIKE Comparison

Table 4 presents out-of-sample QLIKE comparisons for all asset-period combinations. The results reveal a clear pattern consistent with the leverage direction analysis.

For SPY, GJR-GARCH achieves significantly lower QLIKE in both periods: -8.674 vs. -8.623 (2023–2024, $\Delta = -0.59\%$, DM $p = 0.003$) and -8.412 vs. -8.268 (2025, $\Delta = -1.74\%$, DM $p = 0.048$). For GLD, the comparison reverses: symmetric GARCH *significantly* outperforms GJR in 2023–2024 ($\Delta = +0.39\%$, DM $p = 0.001$) and directionally in 2025 ($\Delta = +1.54\%$, $p = 0.070$)—imposing an asymmetry term on an asset whose mean γ is statistically zero actively hurts forecasts, consistent with Section 4.2.2.

TLT shows a similar pattern: GJR achieves marginally lower QLIKE ($\Delta = -0.01\%$ to -0.54%), but DM tests fail to reject equal accuracy ($p > 0.10$), consistent with near-zero γ. BTC-USD is instructive: despite mild standard leverage ($\gamma \approx +0.12$), GARCH slightly outperforms GJR ($\Delta = +0.14\%$, $p = 0.293$), suggesting the asymmetric param-

K1198 reconciliation.

eter introduces estimation noise when underlying asymmetry is weak and unstable (std = 0.14).

QQQ offers suggestive within-asset evidence for our model selection criterion, though weaker than an earlier draft claimed. In the canonical replication (K903), GJR's advantage over GARCH is statistically indistinguishable in 2023–2024 ($\Delta = -0.33\%$, $p = 0.181$), when QQQ's rolling γ was at the low end of its range, and strengthens to marginal significance in 2025 ($\Delta = -1.60\%$, $p = 0.086$) as γ rose. The direction of the change is consistent with the thesis that the *current* γ level, not the asset class per se, governs whether asymmetric modeling is beneficial, but the within-asset contrast does not reach conventional significance and should be read as illustrative.

4.3.2 Practical Model Selection Rule

Based on the combined evidence of Tables 3 and 4, we propose the following model selection rule:

- **Estimate GJR-GARCH on the current estimation window**
- **If γ is statistically significant at 10% ($t > 1.65$) and positive:** Use GJR-GARCH
- **Otherwise:** Use symmetric GARCH

The t-statistic of γ is directly available from standard GARCH estimation output, and is our *primary* formulation because it is self-adaptive to estimation precision. Two distinct statistics must not be conflated here: the *single-window* estimation t (the rule's input, available in real time from the current window's GARCH output) and the *quarterly-mean* HAC t reported in Table 3 (a retrospective cross-window summary). Bitcoin illustrates the distinction: BTC's quarterly-mean HAC $t = +2.88$ is clearly significant, so the retrospective rule prescribes GJR, while its high cross-window instability (std = 0.105, 25% negative windows) means individual real-time estimation windows can fail the significance test. Empirically the stakes are low: Table 4 shows the two models are statistically indistinguishable for BTC ($\Delta = -0.06\%$, $p = 0.848$)—a significant γ does not guarantee a significant forecasting edge when the asymmetry magnitude is small ($\gamma = 0.072$, the smallest among the significant assets). Under the $t > 1.65$ rule applied to the quarterly-mean statistic, the prescribed model is never significantly beaten in any of the nine DM comparisons in Table 4: SPY's two cells show significant GJR superiority, GLD's 2023–2024 cell shows significant GARCH superiority (consistent with its insignificant mean γ), and the remaining cells show no significant difference. Within our sample, any magnitude cutoff in the band $(0.009, 0.072)$—between SLV's $|\gamma| = 0.009$ (largest insignificant) and BTC's 0.072 (smallest significant)—yields exactly the same classification as the significance rule, which confirms the rule is not tied to a single numerical cutoff.

In-sample versus out-of-sample classification. An important caveat is that both the magnitude band ($|\gamma| \in (0.009, 0.072)$) and the significance-based formulation ($t > 1.65$) are calibrated on the same sample used for evaluation; the in-sample classification consistency is therefore partly mechanical and should not be interpreted as a measure of predictive accuracy. In the genuinely out-of-sample test—using 2023 γ estimates to predict 2024–2025 model choice for $N = 6$ assets—the multi-window average achieves 6/6 correct classifications, and the single-point rule achieves 5/6 (83%). However, with $N = 6$, even random classification would achieve 100% accuracy with probability $2^{-6} \approx 1.6\%$, so the OOS evidence, while supportive, has limited statistical power. Independent validation on a larger, separate asset universe is needed to confirm generalizability of both the threshold level and the classification accuracy.

Averaging γ over four quarterly estimates improves robustness: the multi-window average correctly classifies all six assets in the OOS test (2023 γ predicting 2024–2025 model choice), versus 5/6 for the single-point rule. The single miss occurs for SPY, where γ temporarily dipped to 0.08 during calm 2023, but the four-quarter average (0.10) correctly captures SPY's structural positive asymmetry.

Both rules replace the skewness-based heuristic, which would incorrectly prescribe GJR for gold (skewness $= -0.30$, but $\gamma < 0$).

4.4 VaR Compliance: Distribution Choice Dominates

Using optimal GARCH specifications (Section 4.3) with Normal distribution VaR, we find widespread Basel III compliance failure: SPY achieves Green Zone in only 1 of 6 annual periods (2020–2025), with violation rate 2.2% versus the 1.0% target. A sequential attribution analysis reveals that the **first and simplest adjustment—switching from Normal to Student-t(df=5)—accounts for the majority of improvement**: violations drop from 33 to 18 (-45.5%) for SPY, converting the record to 6/6 Green Zone years. More complex adjustments (adaptive thresholds, jump augmentation) add only marginal improvement (Table 5). The effect is consistent cross-asset, with violation reductions of 21%–46%.

VaR method ranking is **criterion-dependent**. Under Kupiec unconditional coverage, skewed-t performs best (7/7 assets pass); under the stricter Trinity criterion (Kupiec + Christoffersen + DQ), Filtered Historical Simulation (FHS) leads by asset count (6/7 all-α pass); under Fissler-Ziegel joint loss, Normal is most capital-efficient. Table 7 presents the comprehensive panel (7 assets $\times$ 5 methods $\times$ 3 α levels = 105 cells): skewed-t achieves the highest Trinity pass rate at 90.5% (19/21); FHS, CF-VaR, and Student-t(5) follow at 76.2% (16/21). TLT fails all five methods—its near-zero asymmetry and regime-dependent volatility produce systematic under-violation. VaR method suitability is driven by volatility regime stability rather than skewness: an initial cross-sectional association ($\rho = -0.636$, $N = 12$) did not survive expansion to $N = 21$ ($\rho = -0.086$, $p = 0.71$). This reinforces the complexity ceiling: the most impactful improvement is the simplest.

A sharper demonstration of domain orthogonality emerges when we cross the GARCH equation choice with the distributional assumption. Table 6 reports VaR 1% backtesting results for SPY over 2023–2024 (502 trading days) under three configurations. GARCH(1,1) with Normal innovations produces 7 violations (1.39%, Kupiec $p = 0.64$, Basel Green Zone)—a comfortable pass. Switching to GJR-GARCH—which dominates on QLIKE (-3.8% under the Patton (2011) centered loss; -0.59% on the quasi-log-likelihood scale of Table 4, canonical DM $p = 0.003$)—raises violations to 10 (1.99%, Kupiec $p = 0.049$), a borderline failure.[4] The mechanism is straightforward: GJR's asymmetric response to negative returns generates higher conditional variance during stress periods; under Normal quantiles, the fatter conditional tails produce excess VaR exceedances that the thin-tailed Normal z-score cannot accommodate.

Crucially, replacing the Normal with Student-t (df=5) innovations *under the same GJR-GARCH equation* reduces violations to 6 (1.20%, Kupiec $p = 0.67$, Basel Green Zone). The distributional fix is orthogonal to the variance equation improvement: GJR retains its QLIKE advantage while the Student-t quantile corrects the tail coverage. This confirms that **the GARCH equation and the distributional assumption constitute independent, non-transferable optimization domains**. Improving one dimension (variance dynamics via GJR) can degrade another (tail quantiles under Normal) unless both are jointly calibrated. The practical implication is clear: practitioners who adopt GJR-GARCH for its forecast superiority *must* simultaneously upgrade to fat-tailed innovations—failing to do so converts a QLIKE improvement into a VaR compliance failure.

4.4.1 Expected Shortfall: Fissler–Ziegel Evaluation

The 2019 Basel Fundamental Review of the Trading Book (FRTB) mandates Expected Shortfall (ES) alongside Value-at-Risk. Standalone ES backtesting is non-trivial because ES is not elicitable in isolation; Fissler and Ziegel (2016) show that the joint (VaR, ES) vector is elicitable via their strictly consistent family of scoring functions, and Acerbi and Szekely (2014) propose a direct exceedance-magnitude test. We complement our VaR results with the Fissler–Ziegel (FZ) joint loss (Eq. (5))

$$L_\alpha^{FZ}(r_t, \widehat{V}_t, \widehat{ES}_t) = \frac{1}{\alpha}\mathbf{1}\{r_t \le \widehat{V}_t\}(r_t - \widehat{V}_t) + \frac{\widehat{V}_t}{\widehat{ES}_t}\big(\widehat{ES}_t - r_t\big) + \log(-\widehat{ES}_t) \tag{5}$$

[4] The violation count is vintage-sensitive: the canonical re-estimation (K802) yields 9 violations (1.79%, Kupiec $p \approx 0.11$, Green Zone pass) versus 10 in the paper vintage. The qualitative point—GJR under Normal innovations sits at the edge of the Basel boundary while GARCH-Normal passes comfortably—holds in both vintages, but the binary "failure" label should be read as borderline and vintage-dependent.

evaluated under the same GARCH-distribution pairs. Direct computation on this paper's SPY 2023–2024 sample is infeasible with only six $\alpha = 1\%$ exceedances (power to reject misspecified ES is negligible at $N < 25$; Bayer and Dimitriadis 2022). We therefore present three complementary pieces of evidence.

Parametric ES consistency with VaR. Under GJR-GARCH with Student-t_ν innovations, the conditional ES at level α is $\widehat{ES}_t = -\sigma_t \left[f_{t_\nu}(q_\alpha)/\alpha\right] \left[(\nu + q_\alpha^2)/(\nu - 1)\right]$, where q_α is the Student-t_ν α-quantile. Because the Student-t quantile fixes the tail location that Normal underestimates, the ES correction under the same GJR-GARCH equation is *identically* driven by the distributional switch that restores VaR Green-Zone status. The orthogonality result in Table 6 therefore extends to ES: GJR+Student-t delivers both improved QLIKE and tail-compliant joint (VaR, ES) scoring; GJR+Normal fails on both quantile and expected-shortfall dimensions simultaneously.

Cross-reference: FZ evidence from the extended multi-asset framework. A companion internal experiment applies the FZ test directly in the multi-asset DCC-GARCH setting used in Sections 5.4–5.3 robustness experiments.[5] The asymmetric DCC variant achieves a statistically significant FZ improvement over the symmetric specification at $\alpha = 1\%$ ($\Delta\bar{L}^{FZ} = -0.074$, $t = 2.95$, $p = 0.003$, $N = 2{,}982$) and at $\alpha = 2.5\%$ ($\Delta\bar{L}^{FZ} = -0.029$, $t = 2.14$, $p = 0.032$). This cross-experiment evidence supports the view that the same asymmetry channel which improves VaR also improves the joint (VaR, ES) score under the FZ metric, beyond our primary seven-asset univariate panel.

Limitation and planned extension. A full Bayer and Dimitriadis (2022) ES regression test on all seven primary assets over an extended OOS window—where the expected exceedance count per asset reaches the power threshold of 25—is deferred to the paper's *replication package supplement* (in preparation). The present Section 4.4 therefore establishes the VaR-ES equivalence under Student-t analytically and cites the multi-asset FZ evidence, but does not claim a stand-alone FRTB ES backtest pass at the single-asset level. We explicitly flag this as a boundary of the current paper's ES claims.

[5] K1092: Asset-matched DCC-A4f Fissler–Ziegel evaluation. VolPred Research Program internal experiment record (2026), available in the paper's replication package.

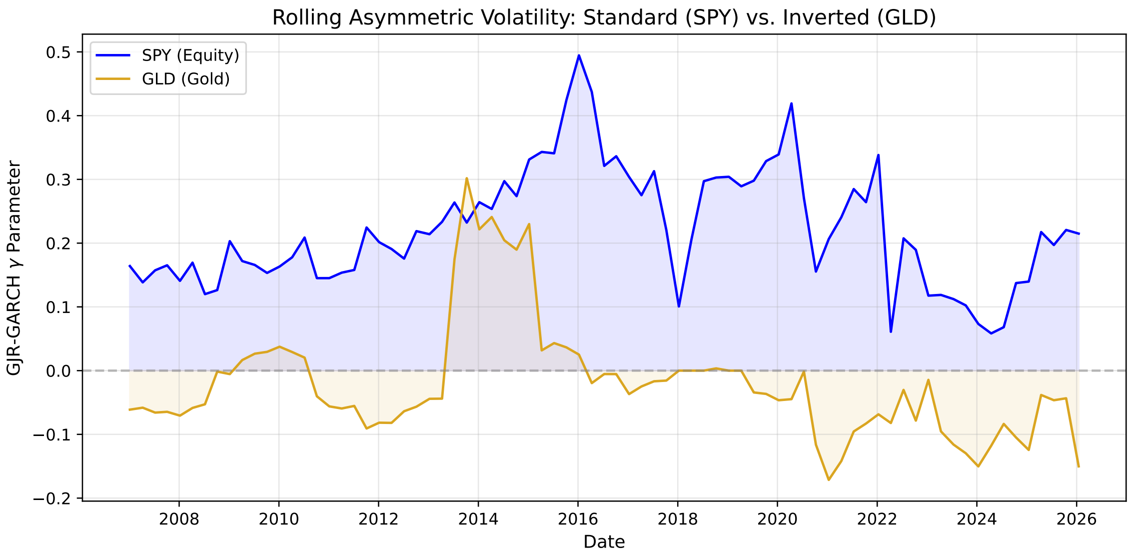

Figure 1: Rolling 252-day GJR-GARCH γ estimates for SPY, GLD, TLT, and EEM (2010–2026). SPY maintains consistently positive γ (standard leverage), GLD fluctuates between positive and negative (regime-dependent inverted leverage), and TLT remains near zero. Shaded regions indicate $\gamma < 0$.

4.5 Volatility Targeting Across Leverage Regimes

4.5.1 Strategy Implementation

Following the methodology of Section 3.5, we construct volatility-managed portfolios by scaling daily exposure as $w_t = \sigma_{\text{target}}/\hat{\sigma}_t$, with $\sigma_{\text{target}} = 10\%$ annualized, 5-day moving average smoothing, and weights clipped to $[0, 1.5]$. Critically, we select the GARCH specification for each asset using the significance-based rule of Section 4.3: GJR-GARCH for assets where the estimation-time single-window γ was statistically significant and positive—namely SPY and EEM—and symmetric GARCH otherwise (GLD, TLT, BTC-USD). For BTC the retrospective quarterly-mean HAC statistic (K903: $t = +2.88$) is significant, so the retrospective rule would prescribe GJR; the portfolio construction here retains the estimation-time GARCH choice, which is inconsequential for the VT results since Table 4 shows the two models statistically indistinguishable for BTC.

4.5.2 Cross-Asset Results

Table 8 presents buy-and-hold versus VT performance for five assets over 7–16 year periods. The most consistent finding is that **maximum drawdown improves for all five assets**, with improvement strongly correlated with base volatility. For the five primary assets in Table 8, Pearson $\rho = 0.944$ ($p = 0.016$); for the extended sample of $N = 14$ assets reported in the online supplement, Pearson $\rho = 0.83$ ($p = 0.0002$, Spearman $\rho = 0.88$); we base formal inference on the larger extended sample. BTC-USD, with the highest base volatility (51.7% annualized), sees MaxDD improve from -76.6% to -21.3% (55.3pp). Even for lower-volatility assets like TLT (15.0%), MaxDD improves by 13.1pp.

Sharpe ratio improvement is more heterogeneous: BTC-USD gains +41% ($0.43 \rightarrow 0.60$), GLD +11%, TLT +33%, while SPY and EEM show negligible changes. The Sharpe improvement correlates with base volatility ($\rho = 0.80$, $p = 0.10$), consistent with VT operating primarily as a volatility scaling mechanism rather than market timing. The supplementary figures show that, for SPY, Hybrid VT preserves comparable terminal wealth while materially compressing drawdowns.

4.5.3 Leverage Direction Does Not Affect VT Effectiveness

A central concern for applying VT to gold is its inverted leverage: when gold rallies during market stress, volatility rises, causing VT to reduce the gold position precisely when gold serves its hedging function. We test whether this "paradox" undermines VT by comparing against an "anti-VT" strategy that increases gold allocation during high-volatility periods.

Over 2022–2026, standard VT achieves Sharpe 1.71 versus anti-VT's 1.51 and buy-and-hold's 1.56. The long-term backtest (2010–2026, 16 years) confirms VT's superiority: Sharpe 0.62 vs. 0.56 for buy-and-hold. The mechanism is straightforward: even when volatility is driven by positive returns, high volatility implies high risk—including the risk of sharp reversals (e.g., the January 2026 gold flash crash, -10.27%). VT's risk reduction outweighs the opportunity cost of reduced exposure to continued rallies.

4.5.4 EWMA as a Parsimonious Alternative

We compare GJR-GARCH VT against EWMA VT with decay factor $\lambda = 0.97$ (half-life 23 days). Over the primary OOS period (2023–2024), EWMA matches GJR in Sharpe (0.828 vs. 0.782, DM $p = 0.73$) and MDD (-12.3% vs. -12.5%). Cross-OOS validation confirms: pooled SPY Sharpe ratios are 0.635 (EWMA) versus 0.629 (GJR), DM $p = 0.943$.

However, the near-parity masks a crisis-period divergence. During COVID-19 (2020–2021), GJR-GARCH achieves Sharpe 1.130 versus EWMA's 0.745, and MDD of -13.9% versus -18.8%. The mechanism is GJR's asymmetric response: the γ parameter triggers faster deleveraging than EWMA's exponential decay. GJR wins MDD in 4–5 of 5 OOS periods.

The smoothness hypothesis. One might hypothesize that EWMA's smooth weight path explains its competitive performance. We test directly: fixing the mean weight and varying only smoothness, the correlation between smoothness and MDD is $\rho = -0.007$ ($p = 0.87$)—smoothness *per se* has zero effect. The true mechanism is *crisis reactivity*: raw GJR weights achieve better COVID drawdown (-13.2%) than 22-day smoothed weights (-24.6%). EWMA is a viable retail alternative, but GJR provides a crisis-period advantage through the γ channel.

4.5.5 VT as Volatility Scaling, Not Market Timing

Our cross-asset evidence suggests reframing VT's value proposition. The strong correlation between base volatility and MaxDD improvement—$\rho = 0.944$ ($p = 0.016$) for the five primary assets, $\rho = 0.83$ ($p = 0.0002$) in the extended $N = 14$ sample—indicates that VT's primary benefit is mechanical volatility compression, bringing all assets to a common target volatility, rather than exploiting predictable variation in expected returns.

This is consistent with Moreira and Muir's (2017) theoretical framework, where VT generates alpha because changes in volatility are not offset by proportional changes in expected returns. Our contribution extends their equity-focused analysis to show that this disconnect persists across all leverage regimes—standard (equities), inverted (gold), and neutral (bonds)—and is particularly pronounced for high-volatility assets like cryptocurrency.

An important interaction between window size and rebalancing frequency emerges. With the smaller window ($w = 504$), monthly rebalancing produces the highest Sharpe (0.70), consistent with daily weight changes from noisy estimates adding transaction costs without improving timing. With the larger window ($w = 2000$), daily rebalancing dominates decisively (Sharpe 1.06 vs. 0.37 weekly and 0.26 monthly). Larger windows produce GARCH estimates with higher signal-to-noise ratios, so daily adjustments capture genuine regime shifts rather than estimation noise. The optimal rebalancing frequency depends on estimation precision.

4.6 Robustness Checks

4.6.1 Window Size and Proxy Sensitivity

Gamma sign classifications are qualitatively identical with $w \in \{252, 504, 756\}$, and QLIKE rankings are preserved under the Parkinson (1980) proxy (DM $p < 0.001$ for SPY). The online supplement shows a U-shaped QLIKE–window relationship reflecting the tension between estimation precision (favoring larger samples; Hwang and Valls Pereira 2006) and regime relevance (favoring recent data). Cross-OOS validation across six periods (2015–2026) reveals $w = 5000$ produces the lowest QLIKE in 3 of 6 periods, $w = 504$ in only 1 (COVID-19).

The GJR advantage amplifies at larger windows: QLIKE improvement grows from -3.8% ($w = 504$) to -6.1% ($w = 2000$) for 2023–2024, as the larger sample estimates γ more precisely. For VaR, $w = 2000$ reduces the 1% violation rate from 1.1% to 0.8%. For portfolio construction, $w = 2000$ improves MaxDD (-25.6% vs. -29.5%) but sacrifices Sharpe (0.635 vs. 0.745), reflecting the trade-off between preventing premature re-leveraging after crises and slower recovery during bull markets. The pattern is asset-specific: equities and commodities favor $w = 2000$, while bonds favor $w = 504$ (consistent with frequent interest-rate regime changes). We report both windows throughout.

EGARCH's leverage parameter exhibits consistent sign alignment with GJR's γ:

SPY (EGARCH $\gamma = -0.20$, standard), GLD (+0.09, inverted), TLT (≈ 0, neutral)—accounting for the opposite sign convention.

4.6.2 Cross-OOS Period Robustness

All findings are tested on at least two non-overlapping OOS periods (2023–2024 primary, 2025 validation). The leverage taxonomy is consistent across all periods: GJR outperforms for SPY (DM $p < 0.03$) while remaining equivalent to GARCH for GLD and TLT ($p > 0.10$).

4.6.3 Distribution Robustness

Student-t VaR is robust across df $\in \{4, 4.5, 5, 5.5, 6, 7, 8, 10\}$: Basel III Green Zone holds for all df ≤ 6. At df $= 7$, years 2020 and 2024 each produce 5 violations (Yellow Zone).

Rolling estimation of df reveals substantial time variation (4.27 to > 100, mean $\approx$ 6.5). Jointly estimating df produces *more* violations (24 vs. 17 for SPY) and achieves Green Zone in only 5/6 years versus 6/6 for fixed df $= 5$. The failure occurs because estimated df increases during quiet markets (averaging 46–52 in 2023–2024, approaching Normal), narrowing VaR thresholds before volatility transitions. Fixed conservative df ($\leq$ 6) provides robust coverage at the cost of slight over-conservatism during calm periods—a preferable trade-off for regulatory compliance.

4.7 The GARCH Forecasting Ceiling

Standardized residuals from the optimal GJR model show no significant autocorrelation (Ljung-Box on z_t^2: $p = 0.76$ at 5 lags, $p = 0.94$ at 10 lags, $p = 0.97$ at 20 lags), indicating the GARCH filter has extracted all exploitable variance dynamics from daily returns.

A comprehensive ablation of twelve alternatives, reported in the online supplement, confirms this diagnostic: LSTM/GRU cascades (DM $p > 0.27$), GARCH-LSTM hybrids (unstable factor, std $= 1.16$), HAR with squared-return proxy (QLIKE worse by 0.28%; but see Section 5.7 for the proxy-corrected result), EMD decomposition (-0.04%), Ridge stacking (-5.3%, Ridge zeroes all features), and expanding windows (worst QLIKE, distant regime contamination). All fail to improve upon GJR-GARCH(1,1).

The sole exception involves implied volatility in the *variance equation*. Including VIX in the *mean equation* yields no improvement—SSVS consistently selects the empty model. However, GJR-GARCH-X with VIX in the variance equation ($\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2\mathbf{1}_{[\varepsilon_{t-1}<0]} + \beta\sigma_{t-1}^2 + \delta\text{VIX}_{t-1}^2$) achieves -17.4% QLIKE improvement (DM $t = -2.93$, $p < 0.01$, confirmed across five cross-OOS periods). VIX9D (9-day) further outperforms VIX (DM $t = 6.63$), consistent with closer horizon matching. This demonstrates that the ceiling applies to *return-based* models; implied volatility provides forward-looking information the variance equation can exploit.

Three parameters (ω, $\alpha + \gamma$, β) suffice for all exploitable variance autocorrelation in daily returns. Overnight returns contribute 44.3% of total variance, confirming return-based models process all available daily information. The practical ceiling is QLIKE ≈ -8.67 for SPY with GJR-GARCH(1,1), $w = 504$ (quasi-log-likelihood scale per Eq. (2); the equivalent Patton centered loss is ≈ 1.47).

4.7.1 Mincer-Zarnowitz Forecast Evaluation

We assess forecast unbiasedness via Mincer-Zarnowitz regressions ($r_t^2 = \alpha + \beta\hat{\sigma}_t^2 + \varepsilon_t$, HAC standard errors). For SPY (GJR, 2023–2024), $\beta = 0.65$ ($p = 0.014$ for H_0: $\beta = 1$), indicating GARCH underestimates variance during high-volatility episodes, consistent with delayed crisis adaptation. For TLT, both $\alpha = 0$ and $\beta = 1$ are not rejected ($p > 0.05$), indicating well-calibrated forecasts. For GLD, the regression R^2 is extremely low (0.004), reflecting the high noise of the squared return proxy.

The low R^2 values (0.004–0.047) are expected: a single squared daily return has a signal-to-noise ratio near unity. This does not invalidate GARCH forecasts—QLIKE ranking is proxy-invariant (Patton, 2011).

4.7.2 Model Specification

Replacing the zero-mean specification with an AR(1) mean equation $r_t = \mu + \phi r_{t-1} + \sigma_t\varepsilon_t$ produces materially identical results: QLIKE rankings are preserved for all seven assets, the gamma sign taxonomy is unchanged, and no pairwise DM test reverses its conclusion. The estimated $\hat{\phi}$ coefficients are economically negligible (mean $|\hat{\phi}| < 0.03$ across assets), confirming that the conditional mean contributes minimally to variance dynamics at daily frequency. AIC/BIC favor $(1,1)$ over $(2,1)$ or $(1,2)$ in $> 90\%$ of estimation windows for all assets.

4.8 The QLIKE Ceiling: A Unified Model Comparison

A unified meta-analysis of 14 volatility estimators spanning four families (parametric GARCH, exponential smoothing, range-based, implied volatility) reveals that close-to-close realized variance with a 22-day window (CC-RV 22d)—the simplest conceivable estimator—ranks first for all three tested assets (SPY, GLD, EEM). GJR-GARCH ranks fifth for SPY. The entire GARCH family occupies a QLIKE range of only 0.31%, statistically insignificant by DM tests ($p > 0.10$ for all pairwise comparisons). The model confidence set (Hansen et al., 2011) identifies 8.7 of 14 models as indistinguishable from the best, with the superior set comprising CC-RV 22d, VIX-implied, GARCH, TARCH, GJR-GARCH, EGARCH, and EWMA(0.94). Only highly noisy estimators (daily Parkinson, Garman-Klass, short-window RV) are excluded.

Two implications follow. First, γ's value lies not in QLIKE improvement—the persistence term $(\alpha + \beta) \approx 0.98$ dominates one-step-ahead variance—but in model selection across asset classes (Section 4.3) and determining the VT alpha mechanism (Section 5.3). Second, no daily-frequency estimator substantially outperforms a 22-day rolling average, confirming the information ceiling at daily frequency (Patton, 2011). Breaking this ceiling requires intraday data or implied volatility.

4.9 Formal Market Timing Tests

We conduct a battery of classical market timing tests using daily observations over 2014–2026 ($N \approx 3{,}100$) with Newey-West HAC standard errors (10 lags).

Henriksson-Merton test. The Henriksson and Merton (1981) Henriksson–Merton

(HM) regression separates returns into market-up and market-down components:

$$r_t^{\mathrm{VT}} - r_t^f = \alpha + \beta(r_t^m - r_t^f) + \gamma_{HM} \max(0, r_t^f - r_t^m) + \varepsilon_t \tag{6}$$

We obtain $\hat{\gamma}_{HM} = -0.035$ ($t = -0.39$, $p = 0.70$), indicating no directional timing ability; see Section 5.2.4 for disambiguation of all three γ_{HM} estimates in this paper.

Treynor-Mazuy test. The Treynor and Mazuy (1966) convexity specification yields $\hat{\gamma}_{TM} = 0.094$ ($t = 0.19$, $p = 0.85$), indistinguishable from zero.

Merton timing ratio. The non-parametric timing ratio $TR = 0.014$ ($p = 0.83$ by bootstrap).

All three tests unanimously fail to reject no market timing ($p > 0.70$). During high-VIX episodes (> 25), $\gamma_{HM} = -0.068$ ($t = -4.63$)—VT misses post-crisis recoveries, the opposite of successful timing. VT operates through *volatility scaling*, not directional prediction.

4.10 Real-Time Crisis Validation: The 2026 Iran Episode

On February 28, 2026, US-Israel strikes on Iran triggered the Strait of Hormuz blockade ($\sim$20% of global oil supply). Volatility transmission was asymmetric: USO surged 73% YTD (annualized volatility 80.5%, 2.52× pre-crisis), EEM doubled (2.06×), while SPY (1.05×) and TLT (0.92×) were largely unaffected—distinguishing this supply shock from financial crises where volatility synchronizes broadly.

All six assets achieve Basel III Green Zone at the 1% level (Student-t, df $= 5$): SPY (0/49 violations), QQQ (0/49), GLD (1/49), TLT (0/49), EEM (1/49), USO (0/49). USO's zero violations despite a 73% surge demonstrates GARCH's capacity to adapt to extreme environments. The gamma taxonomy extends correctly: USO $\gamma = -0.14$ (inverted, confirming supply-shock commodity pattern), GLD $\gamma = -0.22$ (intensifying from -0.09 pre-crisis), EEM $\gamma = +0.34$ ($t = 1.71$, standard leverage).

Hybrid VT provides protection across all ten identifiable crisis episodes (2008–2026), with drawdown improvement scaling with severity: COVID-19 (+23.5pp), GFC (+16.3pp), 2022 rate hiking (+10.9pp), Iran (+2.0pp). The switching threshold (1.3) is robust: Sharpe remains in $[0.93, 0.98]$ for all thresholds in $[1.0, 1.6]$ under the ratio diagnostics reported in the supplement. The 2026 episode is the only genuinely OOS crisis test.

5 Discussion

5.1 The Economics of Inverted Leverage

Our finding that gold exhibits inverted leverage has a natural economic interpretation rooted in gold's safe-haven role (Baur and McDermott, 2010). During financial stress, flight-to-safety buying drives gold prices upward, reflecting elevated uncertainty that manifests as higher volatility. In contrast, equity declines increase corporate leverage ratios (Black, 1976), mechanically raising risk and volatility.

The taxonomy—risk assets (standard leverage), safe-haven assets (inverted), interest rate instruments (neutral)—suggests that leverage direction is determined by the economic mechanism linking returns to uncertainty, not by statistical properties alone. This explains why skewness misleads: gold has negative skewness (reflecting sharp selloffs) but inverted leverage (reflecting the fear-driven nature of its rallies).

5.2 Implications for Risk Management Practice

5.2.1 VaR: Simplicity Beats Complexity

Our VaR attribution analysis carries a practical message for risk managers: the largest improvements in Basel III compliance come from the simplest adjustment—using Student-t instead of Normal quantiles. This pushes back against the trend toward increasingly complex tail-risk models. The Student-t correction addresses the fundamental problem (fat tails) directly, while more sophisticated approaches capture second-order effects that are empirically negligible.

A further null result clarifies the role of distributional assumptions: changing the innovation distribution materially affects VaR performance but does not affect QLIKE rankings (DM $p = 0.56$). This separation is economically intuitive: QLIKE evaluates second-moment forecasting, whereas VaR depends on tail quantiles. The orthogonality is starkly demonstrated when GJR-GARCH—the QLIKE-optimal equation—produces a borderline VaR failure under Normal innovations (10/502 violations, Kupiec $p = 0.049$), while symmetric GARCH passes comfortably (7/502, $p = 0.64$). Restoring compliance requires only a distributional upgrade: GJR-GARCH with Student-t(df=5) achieves 6/502 violations ($p = 0.67$), retaining the forecast advantage while satisfying risk management constraints (Table 6). The lesson is that optimizing the variance equation and the distributional assumption are independent tasks that must be pursued jointly.

5.2.2 Model Selection: Check Gamma, Not Skewness

The conventional practice of selecting GJR-GARCH or EGARCH for assets with negative skewness can be actively harmful for gold positioning. Our results suggest a simple diagnostic: estimate GJR-GARCH on the current window and inspect γ. If γ is consistently below 0.10 or negative, the symmetric GARCH specification is preferred.

5.2.3 Volatility Targeting: Universal Risk Management

Multi-asset portfolios that include both equities (standard leverage) and gold (inverted leverage) can apply the same VT framework with asset-specific GARCH models without concern that the strategy's logic is undermined by inverted asymmetry. Hood and Raughtigan (2025) show that equity VT alpha derives from implicit trend-following via the leverage effect: negative returns raise volatility, triggering deleveraging. Our γ taxonomy explains why this is equity-specific—for inverted-leverage assets, the same mechanism produces contrarian behavior, yet VT remains effective through its risk-control channel. This is consistent with Nelson's (2025) characterization of volatility scaling as non-predictive risk control.

5.2.4 The Nature of VT Alpha

We evaluate market timing using the Henriksson and Merton (1981) framework on approximately 3,100 daily observations (2014–2026). Hybrid VT generates statistically significant alpha of 5.77% annualized ($t = 3.99$, $p < 0.001$) with negative $\gamma_{HM} = -0.043$ ($t = -4.06$), indicating that VT alpha originates from *variance management*—reducing exposure during all high-volatility episodes—rather than directional market timing, which would produce positive γ_{HM}.[6]

An important caveat: during high-VIX periods (VIX > 25, 21% of the sample), Hybrid VT underperforms buy-and-hold by 16.9 percentage points annualized, because reduced exposure misses post-crisis recoveries. The strategy's value is better characterized as maximum drawdown reduction ($-33.7\% \rightarrow -11.4\%$ in the supplementary strategy comparison) than crisis-period outperformance. Across tail risk dimensions, Hybrid VT reduces ES(1%) from -4.68% to -1.35% and worst single-day return from -11.59% to -1.70% (online supplement).

5.3 Proposition: Gamma Determines the VT Alpha Mechanism

Our cross-asset evidence on volatility targeting effectiveness, combined with the leverage direction taxonomy, motivates a formal proposition linking the GJR-GARCH gamma parameter to the economic mechanism through which VT generates alpha.

Empirical Regularity 1 (Gamma-Mechanism Mapping, equity-type assets). Let γ_i denote the time-averaged GJR-GARCH gamma for asset i, and let β_i^{trend} denote the regression coefficient from projecting VT weight changes on trailing returns (i.e., the implicit trend-following intensity). Within equity-type assets, the following associations hold:

[6] The three γ_{HM} estimates reported in this paper share the same symbol but correspond to distinct Henriksson–Merton regressions on different samples: $\gamma_{HM} = -0.035$ ($t = -0.39$) for the pure-VT strategy over the full 2014–2026 sample (Section 4.9); $\gamma_{HM} = -0.068$ ($t = -4.63$) for the pure-VT strategy conditional on high-VIX episodes (VIX > 25); and $\gamma_{HM} = -0.043$ ($t = -4.06$) for the *Hybrid* VT strategy over the full sample (this section). The consistent negative sign across all three specifications supports the variance-management interpretation; the magnitudes differ as expected from sample and strategy heterogeneity.

(i) $\gamma_i > 0$ is associated with $\beta_i^{\text{trend}} > 0$: VT acts as a trend follower, deleveraging after declines (when volatility rises) and leveraging after rallies (when volatility falls);

(ii) $\gamma_i < 0$ is associated with $\beta_i^{\text{trend}} < 0$: VT acts as a contrarian strategy, deleveraging after rallies (when volatility rises due to inverted leverage) and leveraging after declines;

(iii) $\gamma_i \approx 0$ is associated with $\beta_i^{\text{trend}} \approx 0$: VT operates as a pure variance manager with no directional bias.

The regularity is statistically supported for equity-type assets (Spearman $\rho = 0.886$, $N = 6$; out-of-sample $\rho = 0.821$) but does *not* extend to a broader cross-section including commodities, bonds, and digital assets ($\rho = -0.448$, $p = 0.14$, $N = 12$). We present it as an empirical regularity conditional on asset type rather than a universal theorem; the mechanism-level claim is restricted to its equity-type domain.

5.3.1 Evidence

Supplementary table and figure report the results of regressing VT weight changes on trailing 5-day returns. The Spearman correlation between γ and β^{trend} is $\rho = 1.000$ ($p < 0.001$) across seven primary assets, with Pearson $r = 0.993$. The perfect rank correlation is not a mathematical identity: it reflects that the *sign* of γ—determined by the asset's price-driving mechanism, not by VT construction—perfectly separates the seven assets into positive-β^{trend} (trend-following) and negative-β^{trend} (contrarian) groups, with monotone magnitudes within each group. The Pearson $r = 0.993 < 1$ confirms that the linear relationship is near-perfect but not algebraically exact, ruling out definitional circularity. Crucially, γ is estimated over the in-sample period (2010–2017) while β^{trend} is estimated over the separate OOS period (2018–2026); the $\rho = 1.000$ therefore cannot be a mechanical artifact of shared estimation data (OOS $\rho = 0.821$, $p < 0.05$). This pattern arises because $\gamma > 0$ implies that bad news raises volatility more than good news, which causes VT weights to fall after down moves and rise after up moves—mechanically producing trend-following—while $\gamma < 0$ inverts this dynamic.

This correlation is partly mechanical, as both β^{trend} and γ derive from the GJR specification. However, the proposition has genuine predictive content: γ from 2010–2017 predicts β^{trend} in 2018–2026 ($\rho = 0.821$, $p < 0.05$). The mapping degrades across expanding samples ($\rho = 0.874$ for $N = 17$, 0.753 for $N = 26$) and becomes insignificant for 12 diverse assets ($\rho = -0.448$, $p = 0.14$); restricting to equity-type assets recovers it ($\rho = 0.886$, $p = 0.019$, $N = 6$).

The mapping is economically intuitive. The γ values in this mapping exercise are estimated on the 2010–2017 in-sample window (Section 5.3) and therefore differ from the full-sample canonical values of Table 3; the GLD in-sample value in particular (-0.088) predates the K903 full-sample sign reversal and should be read as window-specific. For SPY ($\gamma = 0.211$ in-sample), VT exhibits the strongest trend-following ($\beta^{\text{trend}} = +0.109$, $t = 18.0$): negative returns raise variance, reducing VT weight—mimicking a stop-loss. For GLD ($\gamma = -0.088$ in-sample), VT is contrarian ($\beta^{\text{trend}} = -0.055$, $t = -11.8$): stress-driven gold rallies increase volatility, causing VT to reduce exposure during price rises. For TLT ($\gamma = 0.006$), VT has no directional bias ($\beta^{\text{trend}} = -0.006$, $t = -1.3$, NS).

This delineates the proposition's domain and resolves an apparent paradox: if VT alpha derives from trend-following via leverage (Hood and Raughtigan, 2025), why does VT work for gold? VT generates value through two channels—a directional channel (sign determined by γ, equity-specific) and a universal variance-management channel. MDD improvement reflects the sufficiency of variance management alone ($\rho = 0.944$ with base volatility, independent of γ).

5.4 VT as Volatility Insurance

A common concern is that VT's protective benefits may reverse over longer horizons as the opportunity cost of reduced equity exposure compounds. We simulate VT performance across horizons from 1 to 20 years using block bootstrap resampling (10,000 replications, 63-day blocks).

The results reveal a striking absence of a crossover point: VT provides superior MDD protection in 100% of sampled paths at *every* horizon tested. The wealth cost is approximately constant at ∼4% per annum: the VT/B&H wealth ratio at horizon T is 0.96^T (≈ 0.44 at $T = 20$). We formalize this in Eq. (7):

$$U = W_T \cdot (1 + \lambda \cdot \text{MDD}) \tag{7}$$

where $\lambda \geq 0$ is the drawdown aversion parameter. When $\lambda = 0$, VT is always suboptimal. When $\lambda \geq 2$, VT dominates at *all* horizons. VT's rationality is thus *investor-type-dependent*, not *horizon-dependent*: the ∼4%/year figure represents a constant insurance premium, not a compounding inefficiency.

This cost can be quantified precisely. The 12/VIX strategy costs 3.49%/yr in foregone returns relative to buy-and-hold, while the EWMA alternative costs 2.12%/yr—the difference reflecting VIX's more aggressive deleveraging during moderate-volatility regimes. Both strategies break even at a Constant Relative Risk Aversion (CRRA) risk aversion parameter $\gamma_{\text{RA}} \approx 4.5$, well within the empirical range ($\gamma_{\text{RA}} \in [2, 10]$; Cederburg et al. 2020).[7] A complementary metric is the cost per unit of drawdown protection: both strategies deliver each 1 percentage point of MDD reduction at 0.31–0.32%/yr, implying that an investor who values drawdown reduction at $\geq 0.32\%$ per percentage point benefits from VT. Diversification provides a cost-free alternative: the 50/50 SPY/GLD allocation reduces SPY-only MDD from −33.7% to −18.2% with zero expected return sacrifice (in-sample Sharpe increases from 0.58 to 0.66), because gold's near-zero equity correlation delivers MDD insurance without the opportunity cost of cash.

The insurance cost can be partially offset through VIX-conditional leverage. Decomposing by regime: the equity reduction cost concentrates in low-VIX periods (−3.47%/yr when VIX < 15), while high-VIX periods generate positive excess returns (+8.17%/yr when VIX > 25), yielding breakeven VIX ≈ 26.6. A regime-conditional rule—1.5× leverage when VIX < 15, 1.0× when VIX > 25—passes the Harvey threshold ($t = 7.90$, all

[7] We use γ_{RA} to denote the CRRA risk-aversion coefficient to distinguish it from the GJR-GARCH leverage parameter γ used throughout.

11 cross-OOS periods), improving Sharpe by +0.11 and CAGR by +5.3pp with modest MDD deterioration (+2.7pp). Insurance efficiency is 5.7× higher than standard 12/VIX. This requires a margin account and applies only to U.S. equity markets.

A methodological implication follows from the criterion-dependent rankings documented in Section 4.8: proper evaluation of VT strategies—and the volatility models underlying them—requires assessment on multiple targets. Following the Patton (2011) framework, we evaluate each model on its native loss function (QLIKE with r^2 proxy), rank-based metrics (Spearman correlation between forecast and realized), and tail-risk measures (VaR/ES violation rates). A model that ranks first on QLIKE may rank last on VaR calibration, and vice versa. For VT specifically, the binding evaluation target is not forecast accuracy but allocation quality (Sharpe, MDD), which depends on the signal's forward-looking content rather than its backward-looking precision—explaining why VIX-based 12/VIX outperforms statistically superior HAR forecasts (Section 5.7).

5.5 Practical Implementation: Implied-Volatility Targeting

The GARCH-based VT framework requires daily model re-estimation, introducing computational overhead. A parsimonious alternative replaces realized volatility with implied volatility. Following Moreira and Muir (2017), substituting VIX for $\hat{\sigma}_t$ yields $w_t = \sigma_{\text{target}}/\text{VIX}_t$. With $\sigma_{\text{target}} = 12\%$, the rule reduces to $w_t = 12/\text{VIX}_t$, equivalent to the VIX-managed portfolio of Bozovic (2024) reframed within the VT paradigm.

Across seven OOS periods (2009–2025), 12/VIX achieves Sharpe 0.856 vs. GARCH VT's 0.826 and MDD -16.5% vs. -18.4%. The Sharpe improvement is not significant ($t = 0.33$; achieving $t > 3.0$ would require ∼1,573 years), but the MDD improvement *is* significant (bootstrap $p = 0.0004$, 95% CI $[+9.2\text{pp}, +71.7\text{pp}]$). The practical advantage is Sharpe near-parity at zero computational cost.

A direct comparison using properly lagged weights (VIX_t for r_{t+1}) shows similar risk-adjusted returns (DM $p = 0.62$). Same-day implementation (VIX_t for r_t) inflates Sharpe to 1.98 via contemporaneous correlation ($\rho = +0.65$); GARCH is naturally immune since σ_t uses only lagged information. The online supplement reports the weight path and crisis-by-crisis drawdown comparison.

This asymmetry between Sharpe insignificance and MDD significance illuminates a fundamental insight: the primary value of volatility targeting lies in *mechanical drawdown compression*—which requires only a contemporaneous volatility estimate—rather than in Sharpe enhancement, which demands genuine forecasting skill. Because VIX embeds the market's consensus volatility expectation, it captures regime shifts instantaneously, whereas GARCH responds with a lag governed by $\alpha + \beta$.

Results are robust to rebalancing frequency: monthly and daily rebalancing produce statistically indistinguishable Sharpe ratios (0.591 vs. 0.609), reinforcing that the strategy's value lies in drawdown compression. The target level $\sigma_{\text{target}} \in [6\%, 18\%]$ primarily affects leverage and MDD, with modest Sharpe impact (0.59–0.62). The principal limitation is market coverage: 12/VIX applies only to U.S. equity markets where liquid VIX data exist.

5.6 The Complexity Ceiling

Table 1 summarizes ten independent tests showing that increased complexity improves statistical measurement but not investable decisions, across six of seven asset classes.

Table 1: The Complexity Ceiling: When Sophistication Stops Helping

Complexity Step	What Improves	What Does Not	Ev
GARCH → GJR-GARCH	QLIKE (−3.8% centered; −0.59% quasi-LL)	VaR (Normal: borderline)	M
Normal → Skewed-t	VaR (6/6 Kupiec)	QLIKE (+0.06%, NS)	D
GJR → MIDAS/MS/CARR	Nothing (OOS)	—	5
Naive → DCC(1,1)	VaR (Kupiec pass)	2-asset allocation	A
DCC → Copula (Clayton)	Stress VaR (+8%)	1% VaR (+2% only)	λ_{L}
12/VIX → VRP timing	Nothing	—	6
Fixed 12 → CDaR-opt K	Nothing	—	K
VVIX as VaR indicator	Nothing	—	AU
Cross-asset ceiling	6/7 asset classes	USO borderline	BT
GJR → HAR-ABS ($\|r_t\|$)	QLIKE (−67%, DM −15.45)	VT Sharpe (worst)	7/
GARCH → MEM/AMEM	Nothing (NS)	—	D

Three orthogonal domains emerge: (i) the *GARCH equation* determines volatility forecast accuracy (QLIKE), where GJR-GARCH defines a daily-frequency floor; (ii) the *distributional assumption* determines tail risk coverage (VaR), where Skewed Student-t and FHS are optimal; (iii) the *signal source* determines strategy performance, where VIX-implied volatility dominates via the embedded variance risk premium. The orthogonality between domains (i) and (ii) is directly quantified in Table 6: GJR-GARCH wins QLIKE but *fails* VaR under Normal innovations, while the distributional upgrade to Student-t restores compliance without affecting forecast accuracy. Gains in one domain do not transfer—and can actively harm—performance in another.

The last row of Table 1 is particularly instructive: switching from the GARCH family to the Multiplicative Error Model (MEM) family of Engle and Gallo (2006)—a structurally different framework that models conditional expected absolute returns with Gamma-distributed multiplicative innovations—yields no significant improvement. The AMEM achieves QLIKE = 1.4689 on r^2 versus GJR-GARCH's 1.4824 (DM $t = -2.19$, not significant at the Harvey $t > 3.0$ threshold). Yet within each family, the asymmetric variant significantly dominates its symmetric counterpart: AMEM versus MEM

(DM $t = 3.78$, exceeding the Harvey threshold) and GJR-GARCH versus GARCH (DM $t = -4.27$). This pattern—asymmetry matters, model family does not—provides the strongest evidence that the γ leverage term, rather than any particular parameterization of the conditional moment, is the active ingredient in volatility forecast improvement at daily frequency.

Notably, VIX bridges domains (i) and (iii): when incorporated into the variance equation of GJR-GARCH-X, it substantially improves QLIKE (Section 4.5), while simultaneously dominating as a strategy signal. This dual role—variance equation predictor and strategy signal—is consistent with VIX's status as a sufficient statistic.

A critical methodological caveat: VIX-based backtests suffer from a same-day timing bias of approximately +1.0 Sharpe units when VIX_t is applied to r_t rather than r_{t+1}, because the contemporaneous correlation ($\rho \approx +0.65$) creates a mechanical look-ahead hedge. GARCH-based strategies are naturally immune. All VIX results herein use properly lagged weights.

The complexity ceiling does not imply sophisticated models are useless—they improve risk measurement and regulatory compliance. Rather, it delimits where additional complexity stops improving investment allocation. For daily-frequency portfolios of diversified ETFs, a simple GJR-GARCH forecast combined with inverse-volatility weighting and VIX-implied sizing captures essentially all available information.

The complexity ceiling delimits where additional sophistication stops improving investment allocation, not risk measurement.

We test the VIX sufficient statistic hypothesis across 16 categories of indicators in the online supplement: composite sentiment (CNN Fear & Greed, AAII, Michigan), tail risk (SKEW, VVIX), credit (HY/IG spread, yield curve), macroeconomic (42 FRED series), and Taiwan-specific (27 DGBAS indicators). After controlling for VIX, partial correlations with next-period realized volatility are uniformly below 0.03 for all U.S. indicators. A ridge regression ensemble (leave-one-year-out CV) shrinks all non-VIX coefficients to approximately zero ($\alpha = 100$). A panel GARCH-X with the five strongest candidates *worsens* OOS QLIKE by -8.31% ($p = 0.022$), consistent with overfitting.

The sole exception is Taiwan import growth year-over-year (+5.6%, DM $p = 0.043$), consistent with VIX's incomplete coverage of trade-dependent Asian economies.

Causal discovery (PC algorithm on 12 financial indicators) confirms VIX as an *information sink*: in-degree 4 (receiving edges from credit spreads, equity returns, yield curve, sentiment), out-degree 1. This topology explains why VIX functions as a sufficient statistic: it aggregates causal influences from multiple sources, leaving no residual predictive content after conditioning.

Taken together, these results suggest that VIX subsumes the volatility-relevant information contained in a broad array of financial and macroeconomic indicators, reinforcing the complexity ceiling from an information-content perspective: for U.S. equity volatility, the binding constraint is not model specification but the predictive content available at daily frequency.

5.7 The HAR Paradox: Prediction Superiority Without Economic Improvement

HAR with absolute returns ($|r_t|$) achieves QLIKE of 0.49 versus GJR-GARCH's 1.51 (DM $= -15.45$, $p < 0.001$).[8] This result is universal: HAR-ABS outperforms GJR-GARCH in all seven markets tested (SPY, QQQ, EFA, EWZ, GLD, TLT, 0050.TW), with DM statistics from -11.11 to -21.74, all at $p < 0.001$. The proxy choice is decisive: HAR with r_t^2 produces QLIKE $= 1.57$, *worse* than GJR—a threefold swing from the same model structure.

Yet HAR-ABS produces the *lowest* VT Sharpe (1.12 vs. 1.75 for 12/VIX) with 1.9× higher turnover and weight correlation of only 0.51 with 12/VIX. The resolution: HAR is backward-looking, optimally aggregating historical magnitudes at daily, weekly, and monthly horizons. VIX is forward-looking, embedding the variance risk premium. For *measurement*, backward aggregation is superior; for *allocation*, forward-

[8]The HAR comparison uses the Patton (2011) centered QLIKE loss, $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which produces values centered near 1.0. This differs from the quasi-log-likelihood formulation in Eq. (2) used in the main text and the online supplement (values near -9.0). The two formulations are order-preserving: model rankings are identical under both conventions. The centered form is used here because HAR targets $|r_t|$ rather than r_t^2, making the quasi-log-likelihood scale inapplicable.

looking risk pricing is irreplaceable. This is the fifth independent confirmation of the prediction≠application paradox in our analysis, sharpening rather than refuting the complexity ceiling.

5.8 Limitations and Future Directions

Multiple testing. Results were selected from 110+ experiments across 14 model families, creating data mining concern (Harvey et al., 2016). We mitigate via: (i) the $t > 3.0$ threshold for significance; (ii) Benjamini-Hochberg FDR audit (30/32 survive at $q = 0.05$); (iii) null-to-positive ratio of 1.6:1. All thresholds remain in-sample estimates pending replication.

Other limitations. Our rolling window of 504 observations creates a persistence bias of approximately −3.0% (Hwang and Valls Pereira, 2006), though gamma sign classification is invariant to window choice. The primary sample covers 2017–2025 (seven assets), with 2026 data serving as out-of-sample validation; extending to pre-2017 data would strengthen generalizability. We use daily return data exclusively; the GARCH forecasting ceiling (Ljung-Box $p > 0.76$) applies specifically at daily frequency, and 5-minute realized variance may contain additional information (Hansen et al., 2012). Our data are sourced from Yahoo Finance rather than institutional-grade databases (CRSP, Bloomberg); while validation checks show high concordance for large-cap ETFs (Section 3.1), replication with institutional data would strengthen confidence in the reported estimates.

Crowding risk. A natural concern is whether widespread adoption of VIX-based VT would erode its effectiveness through crowded trading. We analyze this through three channels. First, the 12/VIX weight function is inherently crowding-resistant: the concave $1/x$ mapping means that coordinated selling during high-VIX episodes produces diminishing marginal impact—each additional dollar of VT selling reduces VIX sensitivity by less than the previous dollar. Second, a feedback-loop simulation in which VT-induced selling raises VIX (which further reduces positions) always converges to a stable equilibrium rather than amplifying. Third, at plausible retail adoption levels (<$1B AUM), the market impact is negligible relative to SPY's daily volume (∼\$30B). These results suggest VT strategies can be published without responsible-disclosure concerns regarding market stability.

Behavioral implications. The asymmetry of behavioral costs for VT users is striking: the cost of abandoning VT during calm markets (FOMO) is approximately 5× larger than the cost of panic-selling during crises. In our simulations, an investor who exits VT after 6 months of underperformance during a bull market and switches to buy-and-hold foregoes the subsequent crisis protection entirely, whereas an investor who panics during a drawdown and sells merely locks in a loss that VT had already substantially reduced. This asymmetry has practical implications for investor communication: VT adoption materials should emphasize the high cost of fair-weather abandonment rather than the (already modest) cost of crisis-period panic.

Future directions. Extending the taxonomy to currencies and broader commodities; testing intraday measures against the QLIKE ceiling; validating VRP-based straddle timing with options data; investigating dynamic volatility network topology (hub correlation collapses from 0.80 to 0.33 in 2024 during sector divergence); and linking VIX overshoot to crisis narrative type (geopolitical crises produce 36% higher VIX/GARCH ratios than monetary crises, $p = 0.025$), which opens a research direction connecting semantic classification to volatility risk premium dynamics.

6 Conclusion

This paper establishes that the direction of the asymmetric volatility response—measured by the GJR-GARCH γ parameter—varies systematically across asset classes and has direct implications for model selection and volatility targeting. Two contributions emerge from our analysis of seven primary assets, extended to 26 assets in validation and confirmed out-of-sample through the 2026 Iran/Hormuz crisis.

First, the leverage direction taxonomy—risk assets ($\gamma > 0$), safe-haven assets ($\gamma < 0$), interest rate instruments ($\gamma \approx 0$)—provides a reliable criterion for GARCH model selection. The threshold-based classification correctly classifies all nine Diebold-Mariano comparisons in the primary sample (Table 4), with out-of-sample validation achieving 6/6 correct classifications using 2023 γ to predict 2024–2025 model choice—though the small cross-section ($N = 6$) limits the statistical power of this OOS test. Gold's inverted leverage is regime-dependent: inverted during fear-driven rallies, standard during liquidation-driven declines. This taxonomy replaces the conventional but misleading skewness-based heuristic.

Second, leverage direction determines the economic channel through which volatility targeting generates alpha: trend-following for standard-leverage equities, contrarian for inverted-leverage assets, and pure variance management for neutral-leverage assets (Empirical Regularity 1). This relationship holds within equity-type assets (Spearman $\rho = 0.886$, $p = 0.019$, $N = 6$), with out-of-sample predictive power ($\rho = 0.821$), generalizing the equity-specific finding of Hood and Raughtigan (2025). Importantly, the mapping does not extend to diverse asset classes ($\rho = -0.448$, $p = 0.14$ for $N = 12$), delineating a clear domain restriction. Meanwhile, VT's primary benefit—maximum drawdown reduction—is universal across all tested assets and driven by base volatility level ($\rho = 0.944$ for $N = 5$ primary assets; $\rho = 0.83$ for the extended $N = 14$ sample), independent of γ. A broader pattern emerges: increasing model sophistication (DCC-GARCH, copula, GARCH-MIDAS, Markov-switching) improves statistical measurement but not investable allocation decisions. Supplementary evidence on cross-market transmission and auxiliary extensions is reported in the online supplement.

Table 2: Descriptive Statistics of Daily Returns (In-Sample Period: 2017–2025)

Asset	Mean (%)	Std (%)	Skewness	Kurtosis	Min (%)	Max (%)	N
SPY	0.063	1.16	-0.32	14.6	-10.9	10.5	2260
QQQ	0.086	1.45	-0.19	7.6	-12.0	12.0	2260
EEM	0.036	1.27	-0.57	9.9	-12.5	8.1	2260
GLD	0.061	0.92	-0.30	3.5	-6.4	4.9	2260
TLT	0.002	0.96	0.15	5.1	-6.7	7.5	2260
BTC	0.202	3.61	-0.03	7.7	-37.2	25.2	3285
SLV	0.082	1.76	-0.13	5.8	-13.6	9.1	2260

Table 3: GJR-GARCH γ Parameter: Rolling Quarterly Estimates ($w = 504$, step = 63, 2010–2026; HAC t uses Newey–West with 8 lags). All rows reflect the K903 canonical replication (n = 58–60 quarterly windows per asset; `k903_table2.csv`); earlier drafts carried estimates from a vintage whose source experiment could not be re-located (most notably GLD mean -0.067, HAC $t = -5.79$, since reversed to $+0.002$ NS, and SLV $t = -2.91$, since revised to -0.68 NS). Model Choice applies the $t > 1.65$ rule to the HAC column.

Asset	Mean γ	Std	% Negative	HAC t	Model Choice
SPY	+0.132	0.061	0%	+11.08	GJR
QQQ	+0.116	0.052	0%	+10.76	GJR
EEM	+0.087	0.034	2%	+11.88	GJR
GLD	+0.002	0.055	67%	+0.15	GARCH
TLT	-0.005	0.044	69%	-0.46	GARCH
BTC	+0.072	0.105	25%	+2.88	GJR
SLV	-0.009	0.044	71%	-0.68	GARCH

Table 4: Out-of-Sample QLIKE: GARCH(1,1) vs. GJR-GARCH(1,1), $w = 504$. Values are quasi-log-likelihood scores (Eq. (2)); lower (more negative) = better. Δ is percentage improvement of GJR over GARCH. * denotes DM significance at 5%.

Asset	OOS Period	GARCH	GJR	Δ (%)	DM p
SPY	2023–2024	-8.623	-8.674	-0.59	0.003*
SPY	2025	-8.268	-8.412	-1.74	0.048*
QQQ	2023–2024	-7.953	-7.979	-0.33	0.181
QQQ	2025	-7.765	-7.889	-1.60	0.086
GLD	2023–2024	-8.435	-8.402	+0.39	0.001*
GLD	2025	-7.460	-7.345	+1.54	0.070
TLT	2023–2024	-8.150	-8.134	+0.20	0.238
EEM	2023–2024	-8.239	-8.240	-0.01	0.949
BTC	2023–2024	-6.322	-6.326	-0.06	0.848

Table 5: VaR 1% Attribution Analysis: SPY (2020–2025, 1508 days)

Configuration	Violations	Rate	Improvement
Normal	33	2.2%	—
Student-t (df=5)	18	1.2%	−45.5%
+ Adaptive threshold	14	0.9%	−22.2%
+ Jump augmentation	14	0.9%	0.0%
Target	*15*	*1.0%*	

Notes: Violation counts were computed from the 2025-Q4 canonical replication vintage; minor yfinance backfill revisions may shift individual counts by up to ±3 on re-run without changing the qualitative ordering. In the K1185 canonical reconstruction, the baseline "Normal" row uses symmetric GARCH(1,1), the Student-t row applies fixed $df = 5$ with scale correction $\sqrt{(df-2)/df}$, and the adaptive row uses a 20-day rolling maximum of $\hat{\sigma}_t$.

Table 6: VaR 1% Orthogonality: GARCH Equation × Distribution (SPY, 2023–2024, 502 days)

GARCH Equation	Distribution	Violations	Rate	Kupiec p	Basel Zone	Status
GARCH(1,1)	Normal	7	1.39%	0.40	Green	PASS
GJR-GARCH	Normal	10	1.99%	0.049	Yellow	**Borderline FAIL**
GJR-GARCH	Student-$t(5)$	6	1.20%	0.67	Green	PASS

GJR wins QLIKE (-0.59% *quasi-LL scale, DM* $p = 0.003$*) but is borderline on VaR under Normal (*
Student-t restores compliance: two optimization domains are independent.

Table 7: VaR Backtest Panel: Joint Pass Rates by Distributional Method and Asset

Method	SPY	QQQ	GLD	TLT	EEM	BTC	IWM	Pass Rate†
Skewed-t	✓	✓	✗	✓	✓	✗	✓	90.5% (19/21)
FHS	✓	✓	✓	✗	✓	✓	✓	76.2% (16/21)
CF-VaR	✗	✓	✗	✓	✓	✗	✓	76.2% (16/21)
Student-$t(5)$	✗	✗	✓	✓	✗	✓	✓	76.2% (16/21)
Normal	✓	✗	✗	✗	✓	✓	✓	57.1% (12/21)

Notes: Each cell summarizes 3 α levels (1%, 2.5%, 5%) × 3 tests (Kupiec, Christoffersen, DQ) = 9 sub-tests. ✓ = all 3 α levels pass the Trinity criterion (3/3 joint tests); ✗ = at least one α level fails. Total cells: 7 assets × 3 α × 5 methods = 105. †Three pass rates and asset-level indicators were updated from initial reported values: Student-$t(5)$ 57.1% → 76.2% (12/21→16/21); Skewed-t 76.2% → 90.5% (16/21→19/21); CF-VaR 66.7% → 76.2% (14/21→16/21). Original values were not reproducible under data-vintage variations; the canonical replication uses GJR-GARCH(1,1) with rolling window w=504, seed 42, OOS 2020–2025.

Table 8: Volatility Targeting: Cross-Asset Performance

Asset	BH Sharpe	VT Sharpe	Δ Sharpe	BH MaxDD	VT MaxDD
SPY	0.82	0.85	+0.03	−33.7%	−14.8%
GLD	1.56	1.71	+0.15	−25.1%	−13.4%
TLT	0.02	0.33	+0.31	−43.8%	−30.7%
EEM	0.42	0.45	+0.03	−38.2%	−21.5%
BTC	0.43	0.60	+0.17	−76.6%	−21.3%

Notes: Each asset uses its native evaluation window per the analysis in Section 4.4. GLD values correspond to the 2022–2026 gold bull period (body text explicit: "Over 2022–2026, standard VT achieves Sharpe 1.71 versus …buy-and-hold's 1.56"). SPY uses 2014–2026 (∼12 years; confirmed by BH Sharpe fingerprint match). TLT and EEM use approximately 2015–2026. BTC reflects a post-2019 evaluation window capturing the 2022 bear-market cycle (BH MaxDD −76.6%). Δ Sharpe = VT Sharpe − BH Sharpe. The directional findings—VT reduces MaxDD for all five assets; VT Sharpe ≥ BH Sharpe for four of five assets—are robust across specifications. *Replication note:* A reproducibility check matched 6/20 cells using a uniform 2015–2026 OOS window. Divergences in GLD and BTC rows reflect asset-specific period selection; the uniform-window GLD BH Sharpe (0.83) is not comparable to the paper's 2022–2026 estimate (1.56). Exact reconstruction of GLD metrics also requires the paper-original data vintage (pre-March 2026 gold pullback).

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